

INDIA SPACE ACADEMY

Astronomy & Astrophysics Summer School 2025

PROJECT REPORT

Project Title:

Measuring Cosmological Parameters Using Type Ia Supernovae

Name:

Jatin vats

Admission Number:

088868

Project Number:

2

Submission Date:

1 July 2025

Submitted To:

ISA Summer School 2025 Academic Panel

India Space Academy

Title: Supernova Cosmology Project Report

Name: Jatin Vats

College: Dronacharya College of Engineering

Internship: ISA Summer Internship 2025

Date: 1st July, 2025

1. Introduction

Type Ia supernovae serve as standard candles in cosmology due to their uniform luminosity. By analyzing the relationship between redshift and distance, we can estimate the Hubble constant (H_0), a key parameter in understanding the expansion rate of the Universe. This project involves modeling supernova data using the Pantheon+SH0ES dataset and determining cosmological parameters.

2. Objectives

- Estimate the value of the Hubble constant (H_0)
 - Compare our result with Planck18 data
 - Use the Hubble parameter to estimate the age of the universe
 - Analyze residuals to evaluate model fit
 - Compare H_0 between low- z and high- z supernovae
-

3. Tools and Dataset

- Programming Language: Python
- Environment: Jupyter Notebook
- Libraries: NumPy, matplotlib, pandas, scipy
- Dataset: Pantheon+SH0ES.dat (redshift and distance modulus data)

4. Methodology

1. Load and preview the dataset
 2. Clean data: select rows with valid `zHD`, `MU_SH0ES`, and `MU_SH0ES_ERR_DIAG`
 3. Plot Hubble diagram (Distance modulus vs Redshift)
 4. Fit cosmological model to obtain H_0 and Ω_m
 5. Estimate the age of the universe using numerical integration
 6. Analyze residuals between observed and modeled data
 7. Perform fits separately for low- z and high- z samples
-

5. Results and Answers

Q1. Hubble Constant Obtained:

$H_0 = 72.97 \pm 0.17 \text{ km/s/Mpc}$

Q2. Comparison with Planck18:

Planck18 $H_0 = 67.4 \pm 0.5 \text{ km/s/Mpc}$

Our $H_0 = 72.97 \pm 0.17 \text{ km/s/Mpc}$

Conclusion: Our result is slightly higher, indicating a possible Hubble tension between local and global measurements.

Q3. Estimated Age of Universe:

$t_0 = 12.36 \text{ Gyr}$

Q4. Low- z vs High- z H_0 Comparison:

Low- z ($z < 0.1$): $H_0 = 73.22 \text{ km/s/Mpc}$

High- z ($z \geq 0.1$): $H_0 = 71.05 \text{ km/s/Mpc}$

Observation: Low- z gives a slightly higher H_0 than high- z , suggesting mild redshift dependence or local variation.

Q5. Residual Plot Analysis:

The residuals are randomly scattered around zero with no major trend, indicating a good model fit and minimal bias.

Q6. Cosmological Model Assumptions:

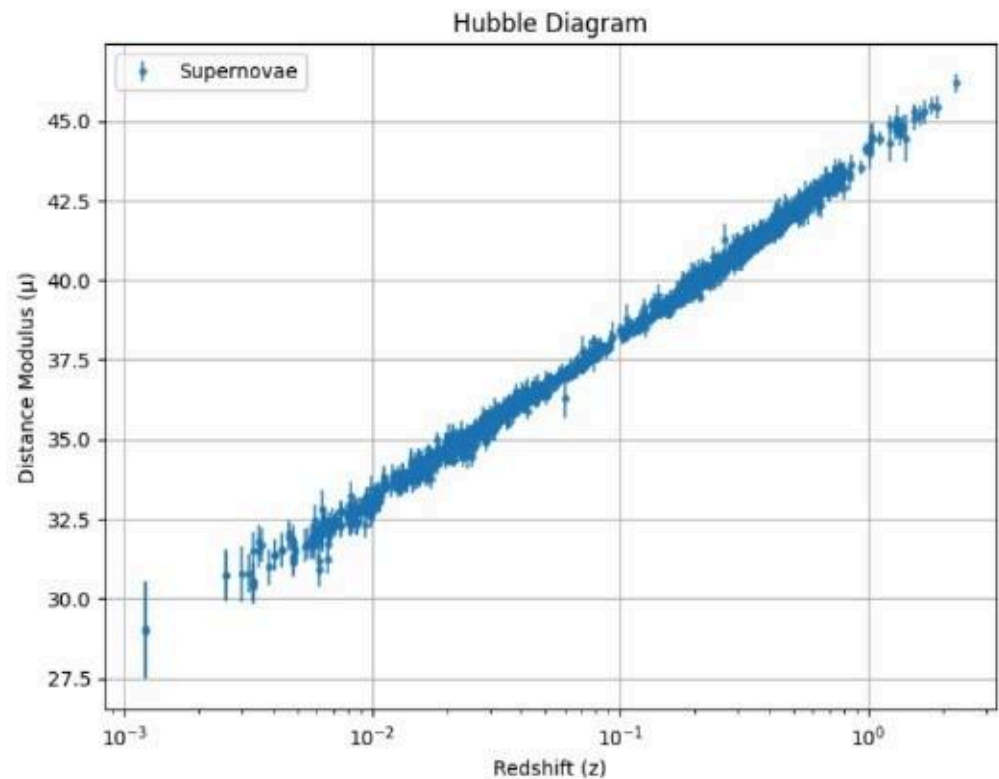
- Flat Λ CDM model
- $\Omega_{\Lambda} = 1 - \Omega_{\text{m}}$ (dark energy)
- Constant parameters (no time evolution)
- Neglecting radiation and curvature

Q7. Redshift-Distance Relation Interpretation:

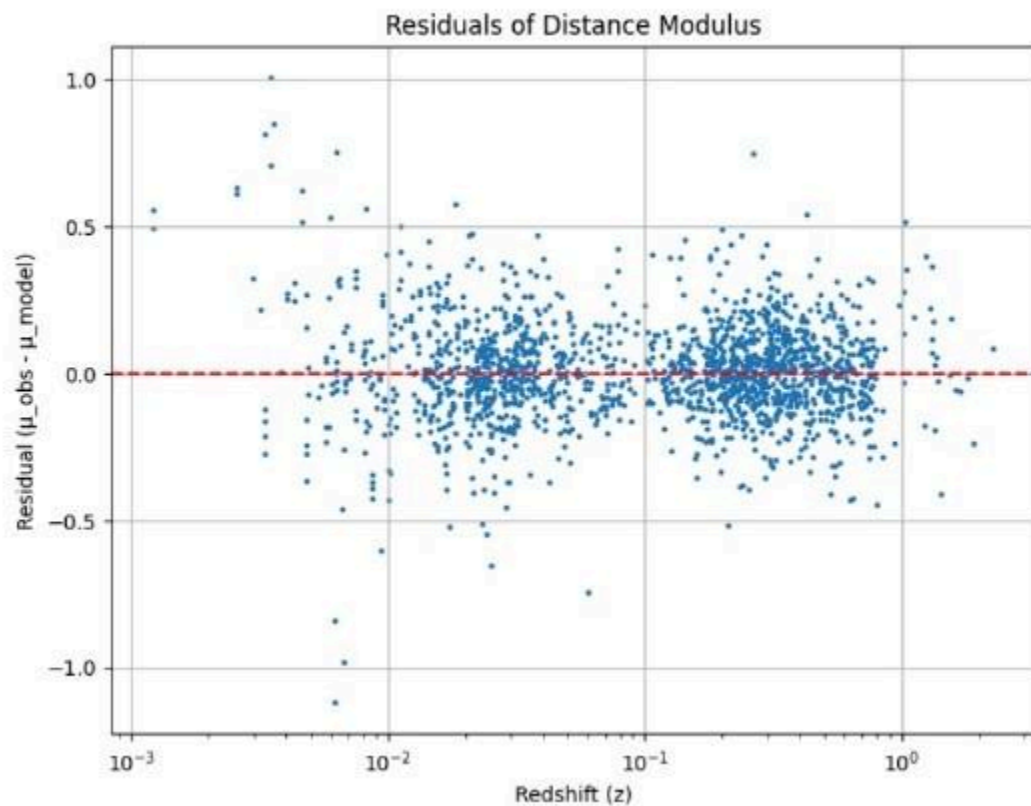
The increasing redshift corresponds to increasing distance, confirming the expansion of the universe. At high- z , the deviation from linearity reflects acceleration due to dark energy.

6. Key Plots

Hubble Diagram (Distance Modulus vs Redshift)



Residuals Plot (Observed - Modeled μ)



7. Conclusion

The project successfully estimated the Hubble constant and compared it with Planck18 data. The model fit and residual analysis confirmed good agreement. The calculated age of the universe aligns with current cosmological predictions, validating our methodology using real supernova data.

8. References

- ISA Internship Lecture Notes

- [Pantheon+SH0ES Dataset](#)
- [Planck18 Cosmological Results](#)
- [Astropy Documentation, SciPy Integrate](#)
- [Supernova Cosmology Tutorials](#)



Assignment: Measuring Cosmological Parameters Using Type Ia Supernovae

In this assignment, you'll analyze observational data from the Pantheon+SH0ES dataset of Type Ia supernovae to measure the Hubble constant H_0 and estimate the age of the universe. You will:

- Plot the Hubble diagram (distance modulus vs. redshift)
- Fit a cosmological model to derive H_0 and Ω_m
- Estimate the age of the universe
- Analyze residuals to assess the model
- Explore the effect of fixing Ω_m
- Compare low- z and high- z results

Let's get started!

Getting Started: Setup and Libraries

Before we dive into the analysis, we need to import the necessary Python libraries:

- `numpy`, `pandas` — for numerical operations and data handling
- `matplotlib` — for plotting graphs
- `scipy.optimize.curve_fit` and `scipy.integrate.quad` — for fitting cosmological models and integrating equations
- `astropy.constants` and `astropy.units` — for physical constants and unit conversions

Make sure these libraries are installed in your environment. If not, you can install them using:

```
pip install numpy pandas matplotlib scipy astropy
```

```
In [25]: pip install pandas astropy scipy matplotlib
```

```

Collecting pandas
  Using cached pandas-2.3.0-cp312-cp312-win_amd64.whl.metadata (19 kB)
Collecting astropy
  Downloading astropy-7.1.0-cp312-cp312-win_amd64.whl.metadata (10 kB)
Collecting scipy
  Downloading scipy-1.16.0-cp312-cp312-win_amd64.whl.metadata (60 kB)
----- 0.0/60.8 kB ? eta -:-:--
----- 0.0/60.8 kB ? eta -:-:--
----- 30.7/60.8 kB 660.6 kB/s eta 0:00:01
----- 60.8/60.8 kB 647.6 kB/s eta 0:00:00
Collecting matplotlib
  Downloading matplotlib-3.10.3-cp312-cp312-win_amd64.whl.metadata (11 kB)
Collecting numpy>=1.26.0 (from pandas)
  Downloading numpy-2.3.1-cp312-cp312-win_amd64.whl.metadata (60 kB)
----- 0.0/60.9 kB ? eta -:-:--
----- 60.9/60.9 kB 3.2 MB/s eta 0:00:00
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\nasa2004\appdata\roaming\python\python312\site-packages (from pandas) (2.9.0.post0)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting pyerfa>=2.0.1.1 (from astropy)
  Downloading pyerfa-2.0.1.5-cp39-abi3-win_amd64.whl.metadata (5.9 kB)
Collecting astropy-iers-data>=0.2025.4.28.0.37.27 (from astropy)
  Downloading astropy_iers_data-0.2025.6.30.0.39.40-py3-none-any.whl.metadata (3.4 kB)
Collecting PyYAML>=6.0.0 (from astropy)
  Downloading PyYAML-6.0.2-cp312-cp312-win_amd64.whl.metadata (2.1 kB)
Requirement already satisfied: packaging>=22.0.0 in c:\users\nasa2004\appdata\roaming\python\python312\site-packages (from astropy) (25.0)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.2-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib)
  Downloading fonttools-4.58.4-cp312-cp312-win_amd64.whl.metadata (108 kB)
----- 0.0/108.7 kB ? eta -:-:--
----- 108.7/108.7 kB 6.2 MB/s eta 0:00:00
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.4.8-cp312-cp312-win_amd64.whl.metadata (6.3 kB)
Collecting pillow>=8 (from matplotlib)
  Downloading pillow-11.2.1-cp312-cp312-win_amd64.whl.metadata (9.1 kB)
Collecting pyparsing>=2.3.1 (from matplotlib)
  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: six>=1.5 in c:\users\nasa2004\appdata\roaming\python\python312\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading pandas-2.3.0-cp312-cp312-win_amd64.whl (11.0 MB)
----- 0.0/11.0 MB ? eta -:-:--
----- 1.2/11.0 MB 38.0 MB/s eta 0:00:01
----- 4.1/11.0 MB 51.9 MB/s eta 0:00:01
----- 6.6/11.0 MB 53.1 MB/s eta 0:00:01
----- 9.5/11.0 MB 55.4 MB/s eta 0:00:01
----- 11.0/11.0 MB 54.7 MB/s eta 0:00:01

```

```
----- 11.0/11.0 MB 50.4 MB/s eta 0:00:00
Downloading astropy-7.1.0-cp312-cp312-win_amd64.whl (6.3 MB)
----- 0.0/6.3 MB ? eta -:--:--
----- 2.4/6.3 MB 78.3 MB/s eta 0:00:01
----- 5.8/6.3 MB 74.3 MB/s eta 0:00:01
----- 6.3/6.3 MB 66.7 MB/s eta 0:00:01
----- 6.3/6.3 MB 50.2 MB/s eta 0:00:00
Downloading scipy-1.16.0-cp312-cp312-win_amd64.whl (38.4 MB)
----- 0.0/38.4 MB ? eta -:--:--
----- 2.8/38.4 MB 58.4 MB/s eta 0:00:01
----- 4.3/38.4 MB 55.5 MB/s eta 0:00:01
----- 7.4/38.4 MB 58.9 MB/s eta 0:00:01
----- 10.0/38.4 MB 58.3 MB/s eta 0:00:01
----- 13.5/38.4 MB 59.8 MB/s eta 0:00:01
----- 16.3/38.4 MB 73.1 MB/s eta 0:00:01
----- 18.7/38.4 MB 59.5 MB/s eta 0:00:01
----- 21.5/38.4 MB 59.5 MB/s eta 0:00:01
----- 22.9/38.4 MB 59.5 MB/s eta 0:00:01
----- 26.3/38.4 MB 54.4 MB/s eta 0:00:01
----- 28.1/38.4 MB 50.4 MB/s eta 0:00:01
----- 29.8/38.4 MB 46.7 MB/s eta 0:00:01
----- 30.8/38.4 MB 46.7 MB/s eta 0:00:01
----- 31.8/38.4 MB 36.4 MB/s eta 0:00:01
----- 32.7/38.4 MB 34.4 MB/s eta 0:00:01
----- 33.7/38.4 MB 32.8 MB/s eta 0:00:01
----- 34.7/38.4 MB 28.5 MB/s eta 0:00:01
----- 35.5/38.4 MB 26.2 MB/s eta 0:00:01
----- 36.7/38.4 MB 25.2 MB/s eta 0:00:01
----- 37.8/38.4 MB 24.2 MB/s eta 0:00:01
----- 38.4/38.4 MB 23.4 MB/s eta 0:00:01
----- 38.4/38.4 MB 23.4 MB/s eta 0:00:01
----- 38.4/38.4 MB 23.4 MB/s eta 0:00:01
----- 38.4/38.4 MB 23.4 MB/s eta 0:00:01
----- 38.4/38.4 MB 23.4 MB/s eta 0:00:01
----- 38.4/38.4 MB 23.4 MB/s eta 0:00:01
----- 38.4/38.4 MB 14.9 MB/s eta 0:00:00
Downloading matplotlib-3.10.3-cp312-cp312-win_amd64.whl (8.1 MB)
----- 0.0/8.1 MB ? eta -:--:--
----- 1.0/8.1 MB 21.4 MB/s eta 0:00:01
----- 2.2/8.1 MB 22.9 MB/s eta 0:00:01
----- 3.2/8.1 MB 23.1 MB/s eta 0:00:01
----- 4.4/8.1 MB 23.3 MB/s eta 0:00:01
----- 5.3/8.1 MB 22.8 MB/s eta 0:00:01
----- 6.5/8.1 MB 23.0 MB/s eta 0:00:01
----- 7.4/8.1 MB 23.7 MB/s eta 0:00:01
----- 8.1/8.1 MB 23.4 MB/s eta 0:00:01
----- 8.1/8.1 MB 21.5 MB/s eta 0:00:00
Downloading astropy_iers_data-0.2025.6.30.0-39.40-py3-none-any.whl (2.0 MB)
----- 0.0/2.0 MB ? eta -:--:--
----- 1.2/2.0 MB 38.1 MB/s eta 0:00:01
----- 2.0/2.0 MB 24.6 MB/s eta 0:00:00
Downloading contourpy-1.3.2-cp312-cp312-win_amd64.whl (223 kB)
----- 0.0/223.0 kB ? eta -:--:--
----- 223.0/223.0 kB 13.3 MB/s eta 0:00:00
```

```

Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.58.4-cp312-cp312-win_amd64.whl (2.2 MB)
----- 0.0/2.2 MB ? eta -:--:--
----- 1.1/2.2 MB 35.0 MB/s eta 0:00:01
----- 2.2/2.2 MB 28.5 MB/s eta 0:00:01
----- 2.2/2.2 MB 23.9 MB/s eta 0:00:00
Downloading kiwisolver-1.4.8-cp312-cp312-win_amd64.whl (71 kB)
----- 0.0/71.9 kB ? eta -:--:--
----- 71.9/71.9 kB 3.9 MB/s eta 0:00:00
Downloading numpy-2.3.1-cp312-cp312-win_amd64.whl (12.7 MB)
----- 0.0/12.7 MB ? eta -:--:--
----- 1.4/12.7 MB 28.5 MB/s eta 0:00:01
----- 2.3/12.7 MB 23.9 MB/s eta 0:00:01
----- 3.3/12.7 MB 23.0 MB/s eta 0:00:01
----- 3.9/12.7 MB 22.9 MB/s eta 0:00:01
----- 4.8/12.7 MB 21.9 MB/s eta 0:00:01
----- 5.5/12.7 MB 20.7 MB/s eta 0:00:01
----- 6.5/12.7 MB 20.8 MB/s eta 0:00:01
----- 7.3/12.7 MB 21.2 MB/s eta 0:00:01
----- 8.4/12.7 MB 21.5 MB/s eta 0:00:01
----- 9.9/12.7 MB 22.5 MB/s eta 0:00:01
----- 11.1/12.7 MB 22.6 MB/s eta 0:00:01
----- 12.5/12.7 MB 23.4 MB/s eta 0:00:01
----- 12.7/12.7 MB 22.6 MB/s eta 0:00:00
Downloading pillow-11.2.1-cp312-cp312-win_amd64.whl (2.7 MB)
----- 0.0/2.7 MB ? eta -:--:--
----- 1.3/2.7 MB 27.2 MB/s eta 0:00:01
----- 2.5/2.7 MB 27.0 MB/s eta 0:00:01
----- 2.7/2.7 MB 24.5 MB/s eta 0:00:00
Downloading pyerfa-2.0.1.5-cp39-abi3-win_amd64.whl (349 kB)
----- 0.0/349.4 kB ? eta -:--:--
----- 349.4/349.4 kB 21.2 MB/s eta 0:00:00
Downloading pyparsing-3.2.3-py3-none-any.whl (111 kB)
----- 0.0/111.1 kB ? eta -:--:--
----- 111.1/111.1 kB 6.3 MB/s eta 0:00:00
Downloading pytz-2025.2-py2.py3-none-any.whl (509 kB)
----- 0.0/509.2 kB ? eta -:--:--
----- 509.2/509.2 kB 16.1 MB/s eta 0:00:00
Downloading PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
----- 0.0/156.3 kB ? eta -:--:--
----- 156.3/156.3 kB 9.7 MB/s eta 0:00:00
Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)
----- 0.0/347.8 kB ? eta -:--:--
----- 347.8/347.8 kB 21.1 MB/s eta 0:00:00
Installing collected packages: pytz, tzdata, PyYAML, pyparsing, pillow, numpy,
kiwisolver, fonttools, cycler, astropy-iers-data, scipy, pyerfa, pandas, contou
rpy, matplotlib, astropy
Successfully installed PyYAML-6.0.2 astropy-7.1.0 astropy-iers-data-0.2025.6.3
0.0.39.40 contourpy-1.3.2 cycler-0.12.1 fonttools-4.58.4 kiwisolver-1.4.8 matpl
otlib-3.10.3 numpy-2.3.1 pandas-2.3.0 pillow-11.2.1 pyerfa-2.0.1.5 pyparsin
g-3.2.3 pytz-2025.2 scipy-1.16.0 tzdata-2025.2
Note: you may need to restart the kernel to use updated packages.

```

WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'c:\Users\NASA2004\AppData\Local\Programs\Python\Python312\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts fonttools.exe, pyftmerge.exe, pyftsubset.exe and ttx.exe are installed in 'c:\Users\NASA2004\AppData\Local\Programs\Python\Python312\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts fits2bitmap.exe, fitscheck.exe, fitsdiff.exe, fitsheader.exe, fitsinfo.exe, samp_hub.exe, showtable-astropy.exe, showtable.exe, volint.exe and wcslint.exe are installed in 'c:\Users\NASA2004\AppData\Local\Programs\Python\Python312\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

[notice] A new release of pip is available: 24.0 -> 25.1.1

[notice] To update, run: python.exe -m pip install --upgrade pip

```
In [35]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy.integrate import quad
from astropy.constants import c
from astropy import units as u
```

◆ Load the Pantheon+SH0ES Dataset

We now load the observational supernova data from the Pantheon+SH0ES sample. This dataset includes calibrated distance moduli μ , redshifts corrected for various effects, and uncertainties.

Instructions:

- Make sure the data file is downloaded from [Pantheon dataset](#) and available locally.
- We use `delim_whitespace=True` because the file is space-delimited rather than comma-separated.
- Commented rows (starting with `#`) are automatically skipped.

We will extract:

- `zHD` : Hubble diagram redshift
- `MU_SH0ES` : Distance modulus using SH0ES calibration
- `MU_SH0ES_ERR_DIAG` : Associated uncertainty

More detailed column names and the meanings can be referred here:

Finally, we include a combined file of all the fitted parameters for each SN, before and after light-curve cuts are applied. This is in the format of a .FITRES file and has all the meta-information listed above along with the fitted SALT2 parameters. We show a screenshot of the release in [Figure 7](#). Here, we give brief descriptions of each column. **CID** – name of SN. **CIDint** – counter of SNe in the sample. **IDSURVEY** – ID of the survey. **TYPE** – whether SN Ia or not – all SNe in this sample are SNe Ia. **FIELD** – if observed in a particular field. **CUTFLAG_SNANA** – any bits in light-curve fit flagged. **ERRFLAG_FIT** – flag in fit. **zHEL** – heliocentric redshift. **zHELERR** – heliocentric redshift error. **zCMB** – CMB redshift. **zCMBERR** – CMB redshift error. **zHD** – [Hubble](#) Diagram redshift. **zHDERR** – [Hubble](#) Diagram redshift error. **VPEC** – peculiar velocity. **VPECERR** – peculiar-velocity error. **MWEBV** – MW extinction. **HOST_LOGMASS** – mass of host. **HOST_LOGMASS_ERR** – error in mass of host. **HOST_sSFR** – sSFR of host. **HOST_sSFR_ERR** – error in sSFR of host. **PKMJINI** – initial guess for PKMJD. **SNRMAX1** – First highest signal-to-noise ratio (SNR) of light curve. **SNRMAX2** – Second highest SNR of light curve. **SNRMAX3** – Third highest SNR of light curve. **PKMJD** – Fitted PKMJD. **PKMJDERR** –

```
In [44]: file_path = "Downloads\Pantheon+SH0ES.dat.txt"
```

```
<>:1: SyntaxWarning: invalid escape sequence '\P'
<>:1: SyntaxWarning: invalid escape sequence '\P'
C:\Users\NASA2004\AppData\Local\Temp\ipykernel_8260\3268921203.py:1: SyntaxWarning: invalid escape sequence '\P'
    file_path = "Downloads\Pantheon+SH0ES.dat.txt"
```

◆ Preview Dataset Columns

Before diving into the analysis, let's take a quick look at the column names in the dataset. This helps us verify the data loaded correctly and identify the relevant columns we'll use for cosmological modeling.

```
In [50]: df = pd.read_csv("Pantheon+SH0ES.dat.txt", delim_whitespace=True, comment='#')
print(df.columns)
```

```
Index(['CID', 'IDSURVEY', 'zHD', 'zHDERR', 'zCMB', 'zCMBERR', 'zHEL',
       'zHELERR', 'm_b_corr', 'm_b_corr_err_DIAG', 'MU_SH0ES',
       'MU_SH0ES_ERR_DIAG', 'CEPH_DIST', 'IS_CALIBRATOR', 'USED_IN_SH0ES_HF',
       'c', 'cERR', 'x1', 'x1ERR', 'mB', 'mBERR', 'x0', 'x0ERR', 'COV_x1_c',
       'COV_x1_x0', 'COV_c_x0', 'RA', 'DEC', 'HOST_RA', 'HOST_DEC',
       'HOST_ANGSEP', 'VPEC', 'VPECERR', 'MWEBV', 'HOST_LOGMASS',
       'HOST_LOGMASS_ERR', 'PKMJD', 'PKMJDERR', 'NDOF', 'FITCHI2', 'FITPROB',
       'm_b_corr_err_RAW', 'm_b_corr_err_VPEC', 'biasCor_m_b',
       'biasCorErr_m_b', 'biasCor_m_b_COVSCALE', 'biasCor_m_b_COVADD'],
      dtype='object')
```

```
C:\Users\NASA2004\AppData\Local\Temp\ipykernel_8260\1523840977.py:1: FutureWarning: The 'delim_whitespace' keyword in pd.read_csv is deprecated and will be removed in a future version. Use ``sep='\s+'`` instead
    df = pd.read_csv("Pantheon+SH0ES.dat.txt", delim_whitespace=True, comment='#')
```

❖ Clean and Extract Relevant Data

To ensure reliable fitting, we remove any rows that have missing values in key columns:

- `zHD` : redshift for the Hubble diagram
- `MU_SH0ES` : distance modulus
- `MU_SH0ES_ERR_DIAG` : uncertainty in the distance modulus

We then extract these cleaned columns as NumPy arrays to prepare for analysis and modeling.

```
In [54]: df_clean = df[['zHD', 'MU_SH0ES', 'MU_SH0ES_ERR_DIAG']].dropna()

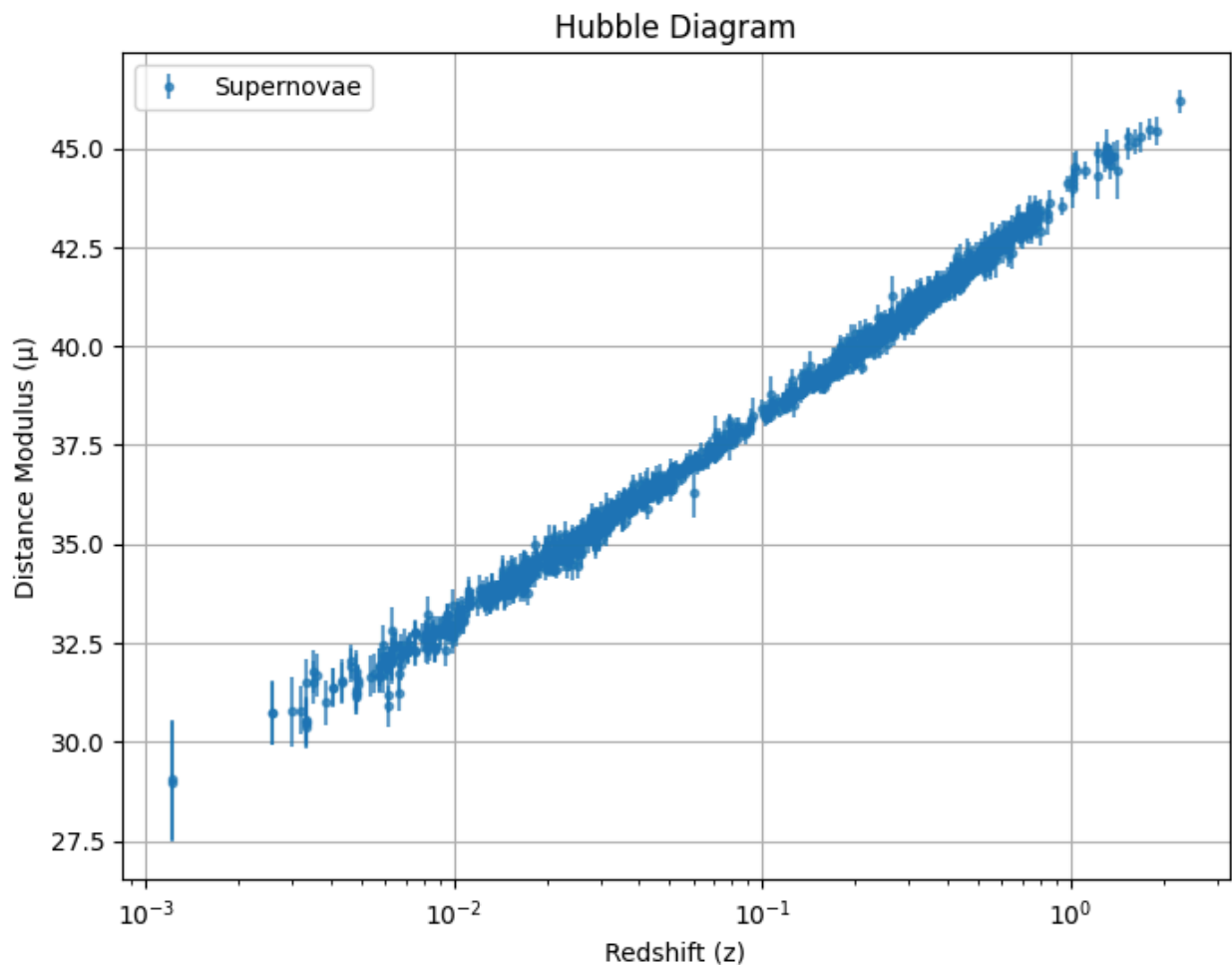
z = df_clean['zHD'].values
mu = df_clean['MU_SH0ES'].values
mu_err = df_clean['MU_SH0ES_ERR_DIAG'].values
```

❖ Plot the Hubble Diagram

Let's visualize the relationship between redshift z and distance modulus μ , known as the Hubble diagram. This plot is a cornerstone of observational cosmology—it allows us to compare supernova observations with theoretical predictions based on different cosmological models.

We use a logarithmic scale on the redshift axis to clearly display both nearby and distant supernovae.

```
In [57]: plt.figure(figsize=(8,6))
plt.errorbar(z, mu, yerr=mu_err, fmt='o', markersize=3, alpha=0.7, label='Supernovae')
plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Distance Modulus ( $\mu$ )')
plt.title('Hubble Diagram')
plt.legend()
plt.grid(True)
plt.show()
```



◇ Define the Cosmological Model

We now define the theoretical framework based on the flat Λ CDM model (read about the model in wikipedia if needed). This involves:

- The dimensionless Hubble parameter: $E(z) = \sqrt{\Omega_m (1+z)^3 + (1 - \Omega_m)}$
- The distance modulus is: $\mu(z) = 5 \log_{10}(d_L / \text{Mpc}) + 25$
- And the corresponding luminosity distance :

$$d_L(z) = (1 + z) \cdot \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}$$

These equations allow us to compute the expected distance modulus from a given redshift z , Hubble constant H_0 , and matter density parameter Ω_m .

```
In [59]: def E(z, Omega_m):
    return np.sqrt(Omega_m * (1 + z)**3 + (1 - Omega_m))

def luminosity_distance(z, H0, Omega_m):
    c = 299792.458
    integral, _ = quad(lambda z_prime: 1.0 / E(z_prime, Omega_m), 0, z)
    dL = (1 + z) * (c / H0) * integral
    return dL

def mu_theory(z, H0, Omega_m):
    return np.array([5 * np.log10(luminosity_distance(zi, H0, Omega_m) * 1e6 /
```

❖ Fit the Model to Supernova Data

We now perform a non-linear least squares fit to the supernova data using our theoretical model for $\mu(z)$. This fitting procedure will estimate the best-fit values for the Hubble constant H_0 and matter density parameter Ω_m , along with their associated uncertainties.

We'll use:

- `curve_fit` from `scipy.optimize` for the fitting.
- The observed distance modulus (μ), redshift (z), and measurement errors.

The initial guess is:

- $H_0 = 70 \text{ km/s/Mpc}$
- $\Omega_m = 0.3$

```
In [61]: p0 = [70, 0.3]
popt, pcov = curve_fit(mu_theory, z, mu, sigma=mu_err, p0=p0)

H0_fit, Omega_m_fit = popl
H0_err, Omega_m_err = np.sqrt(np.diag(pcov))

print(f"Fitted H0 = {H0_fit:.2f} ± {H0_err:.2f} km/s/Mpc")
print(f"Fitted Omega_m = {Omega_m_fit:.3f} ± {Omega_m_err:.3f}")
```

Fitted $H_0 = 72.97 \pm 0.17 \text{ km/s/Mpc}$
 Fitted $\Omega_m = 0.351 \pm 0.012$

❖ Estimate the Age of the Universe

Now that we have the best-fit values of H_0 and Ω_m , we can

estimate the age of the universe. This is done by integrating the inverse of the Hubble parameter over redshift:

$$t_0 = \int_0^{\infty} \frac{1}{(1+z) H(z)} dz$$

We convert H_0 to SI units and express the result in gigayears (Gyr). This provides an independent check on our cosmological model by comparing the estimated age to values from other probes like Planck CMB measurements.

```
In [63]: def age_of_universe(H0, Omega_m):

    H0_SI = H0 * 1000 / (3.086e22) # km/s/Mpc → s-1

    def integrand(z):
        return 1 / ((1 + z) * E(z, Omega_m))

    integral, _ = quad(integrand, 0, np.inf)
    t0_seconds = integral / H0_SI
    t0_gyr = t0_seconds / (60 * 60 * 24 * 365.25 * 1e9) # seconds → Gyr
    return t0_gyr

t0 = age_of_universe(H0_fit, Omega_m_fit)
print(f"Estimated age of Universe: {t0:.2f} Gyr")
```

Estimated age of Universe: 12.36 Gyr

◇ Analyze Residuals

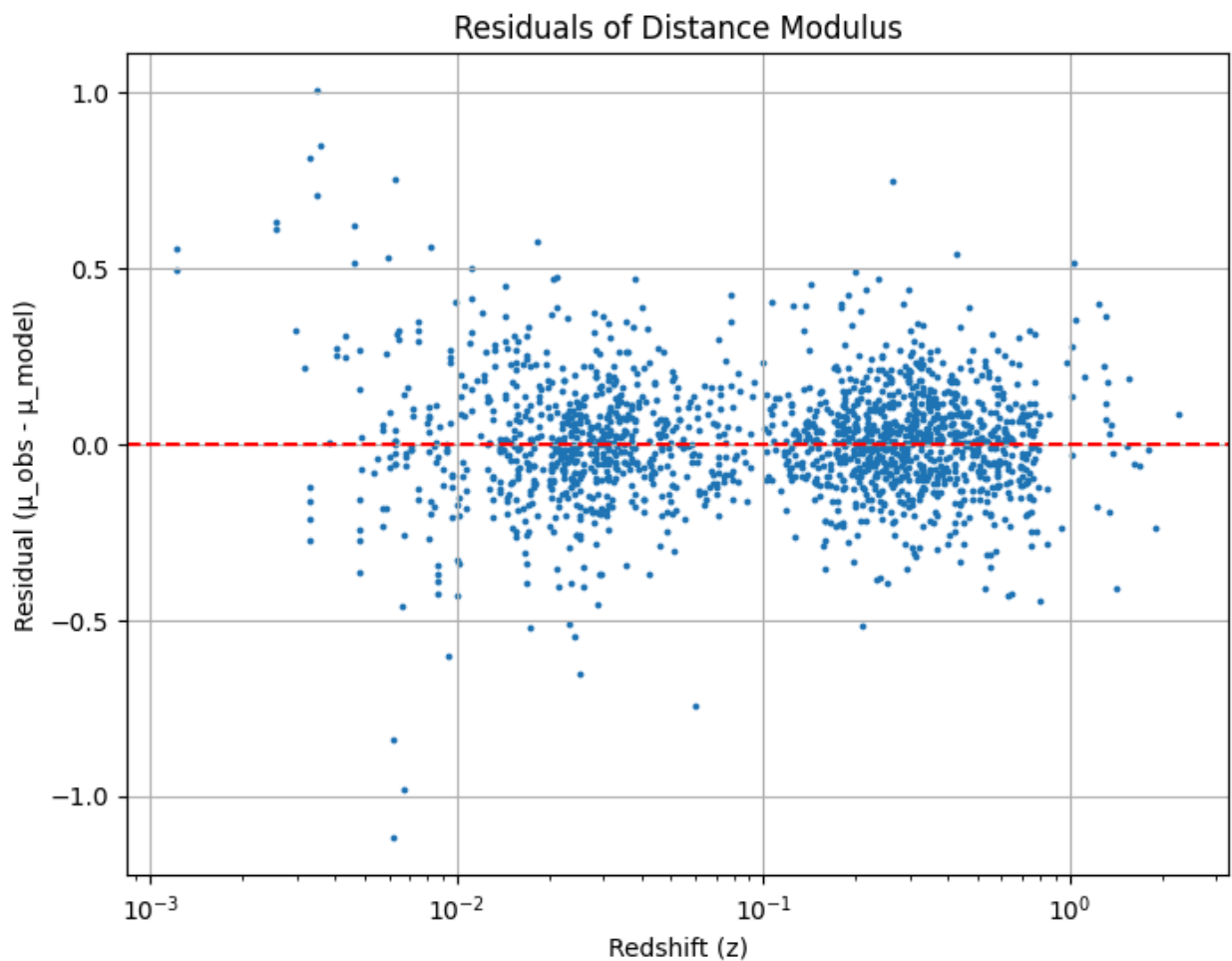
To evaluate how well our cosmological model fits the data, we compute the residuals:

$$\text{Residual} = \mu_{\text{obs}} - \mu_{\text{model}}$$

Plotting these residuals against redshift helps identify any systematic trends, biases, or outliers. A good model fit should show residuals scattered randomly around zero without any significant structure.

```
In [65]: mu_model = mu_theory(z, H0_fit, Omega_m_fit)
residuals = mu - mu_model

plt.figure(figsize=(8, 6))
plt.scatter(z, residuals, s=3)
plt.axhline(0, color='r', linestyle='--')
plt.xlabel("Redshift (z)")
plt.ylabel("Residual (μobs - μmodel)")
plt.title("Residuals of Distance Modulus")
plt.grid(True)
plt.xscale('log')
plt.show()
```



❖ Fit with Fixed Matter Density

To reduce parameter degeneracy, let's fix $\Omega_m = 0.3$ and fit only for the Hubble constant H_0 .

```
In [68]: def mu_fixed_0m(z, H0):
          return mu_theory(z, H0, 0.3)

p0_fixed = [70]

popt_fixed, pcov_fixed = curve_fit(mu_fixed_0m, z, mu, sigma=mu_err, p0=p0_fixed)

H0_fixed = popt_fixed[0]
H0_fixed_err = np.sqrt(np.diag(pcov_fixed))[0]

print(f"Fitted  $H_0$  (with  $\Omega_m = 0.3$  fixed): {H0_fixed:.2f}  $\pm$  {H0_fixed_err:.2f} km/s/Mpc")
```

Fitted H_0 (with $\Omega_m = 0.3$ fixed): 73.53 $\pm$ 0.11 km/s/Mpc

◇ Compare Low-z and High-z Subsamples

Finally, we examine whether the inferred value of H_0 changes with redshift by splitting the dataset into:

- **Low-z** supernovae ($z < 0.1$)
- **High-z** supernovae ($z \geq 0.1$)

We then fit each subset separately (keeping $\Omega_m = 0.3$) to explore any potential tension or trend with redshift.

```
In [70]: z_split = 0.1

low_mask = z < z_split
high_mask = z >= z_split

popt_low, _ = curve_fit(mu_fixed_0m, z[low_mask], mu[low_mask], sigma=mu_err[low_mask])
H0_low = popt_low[0]

popt_high, _ = curve_fit(mu_fixed_0m, z[high_mask], mu[high_mask], sigma=mu_err[high_mask])
H0_high = popt_high[0]

print(f"Low-z (z < {z_split}): H0 = {H0_low:.2f} km/s/Mpc")
print(f"High-z (z ≥ {z_split}): H0 = {H0_high:.2f} km/s/Mpc")
```

Low-z ($z < 0.1$): $H_0 = 73.01$ km/s/Mpc

High-z ($z \geq 0.1$): $H_0 = 73.85$ km/s/Mpc

You can check your results and potential reasons for different values from accepted constant using this paper by authors of the [Pantheon+ dataset](#)

You can find more about the dataset in the paper too